



How GAIA works

Accountability, records, and participation

A collaboration of the University of Washington, the University of Alaska Fairbanks, and the EarthScope Consortium, under three linked NSF awards:

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GEOHAZARD
CENTER



COLLEGE OF THE ENVIRONMENT
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Why we publish this

Projects usually keep their governance internal. We are publishing ours.

It can be checked

A written commitment can be held against us. "Open" on its own cannot.

It can be reused

Other distributed projects face exactly these questions.

It is more honest

A project that shows the targets it has missed is easier to believe.

What we committed to deliver

From the funded proposals. Year 1 is shaded.

	Y1	Y2	Y3	Y4	Y5
Workflow templates with passing tests	3	6	10	14	18
Container images	3	8	15	25	35
Datasets archived with a DOI	5	10	20	50	100
Versioned model cards	1	5	8	12	15
Books and workshops, per year	2	3	4	6	7

Collected weekly and automatically from public sources.

What we committed to demonstrate

Software that no one installs is indistinguishable from software that does not exist.

	Y1	Y3	Y5
Downloads and container pulls, per year	500	5,000	20,000
Agents derived from ours	2	29	100
Publications that used GAIA	0	10	30
Institutions represented	20	50	100
Disciplines represented	3	5	7
Gain in skills, post minus pre	+20%	+40%	+40%

Measurement is for steering, not for reporting.

Collected once a year, these numbers would only describe what already happened.
Collected weekly, they can still change what we do.

Two conventions keep them honest. A figure we cannot collect is recorded as unavailable, never as zero. Figures below target stay on the public page.

When a target is missed

A measurement with no response attached is decoration.

Weekly

The dashboard updates. No action is expected.

Monthly

The thrust lead proposes a correction, or accepts the shortfall and explains why.

Quarterly

Persistent shortfalls go to the Pls: reallocate, revise the target, or consult the programme officer.

Annually

The report states what was missed and what followed.

What we write down

Decisions

Each decision that constrains later choices receives a number, its reasoning, and the alternatives considered.

So a settled question is not reopened from memory.

Meeting notes

Drafted from the recording, attributed by role, kept in a private repository.

So the project's memory does not rest with one person.

Weekly summary

Drafted by an agent, then corrected and approved by a colleague before publication.

So the record survives without recording daily conversation.

What is public

Immediately

The coordination repository, decisions, notes, and the dashboard — including figures below target.

Monthly, prepared

A short public seminar: two five-minute talks and one demonstration. Taking part is voluntary.

Kept internal

Working meetings, held privately. Personnel and budget are never recorded.

We do not record working meetings publicly.

Public recording discourages the tentative idea, the admission of confusion, and the negative result — the substance of research conversation. Discussion then moves to private messages, where nothing is preserved at all.

Openness is better served by publishing what we produce than by observing how we work.

What the GitHub organisation holds

How the project runs

The website and book · the private notes repository · the metrics observatory

Software we maintain

Workflow templates · container images · the evaluation library · research agents

Software we only point to

A curated index of community tools, carrying no maintenance commitment

The science

Use-case repositories and the chapters that describe them

How outside work joins GAIA

Five levels. Each is a larger commitment by us, so each is agreed rather than assumed.

- | | | |
|---|----------------------|--|
| 0 | Listed | A line in the index. Anyone may propose one; we keep it curated. |
| 1 | Containerised | We publish a tested image. The software stays with its authors. |
| 2 | Demonstrated | A template repository shows a complete workflow, and we keep it working. |
| 3 | Benchmarked | It becomes a task or baseline in the evaluation hub. |
| 4 | Adopted | The repository moves to us. Rare, and only with a named maintainer. |

What we learned from SCOPED

Its container registry is the closest precedent we have.

A small base image, with thin layers above it

Separate layers added what HPC or cloud work required. We inherit this.

One repository per tool

NoisePy, MTUQ and ELEP each had their own, maintained by people who knew the software.

Pinned versions, not containers, gave reproducibility

Workshop material still runs because versions were fixed.

One tension it never settled

HPC wants small images; teaching wants notebooks and data. Two tags, not one compromise.

METHOD

Agents draft. We approve.

Agents prepare notes, summaries, and metric commentary. None of it is published unread.

Every drafted document names the model, the date, and the colleague who approved it.

Without that line, a claim about working this way cannot be examined.

Nothing external is sent unread

Reports and correspondence are approved by a person.

References are verified

No citation is published without being checked.

We evaluate our own agents

In the same framework as the scientific ones.

What this asks of you

Post a short status

Weekly if GAIA funds a quarter of your time; monthly otherwise.

Use the acknowledgement

One agreed wording, in papers and in every repository.

Record contributions as you go

Noted while the work happens, not reconstructed at submission.

Give a five-minute talk

Once or twice a year. Results that did not work are welcome.

Say early when something is behind

Revising a plan is ordinary. Discovering it late is not.

What we are asking you to agree

Each has a written note. If no one objects, the proposal stands.

	Proposal	When
1	A single wording for acknowledging the awards	Today
2	Slack for discussion, GitHub for the record	Today
3	Two standing meetings, and a rule for retiring them	Today
4	One project account, not two	Today
5	Authorship, and the citation of software	September
6	What we record, and what we publish	September

The last two are left open for comment rather than settled today.

This week

Complete the intake form · join the organisation and Slack

Add your talks to the shared calendar · comment on the six notes

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